ModelIt!

Teacher's Guide

Why Are Species Going Extinct So Fast?

Modeling Biodiversity Loss, Habitat Fragmentation, and Extinction Cascades

Quick Reference

Grade Level	Grade 12L2
NGSS Standard	HS-LS2-7
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS2-7

Design, evaluate, and refine a solution for reducing the impacts of human activities on the environment and biodiversity.

Three-Dimensional Learning

Science Practice	**Designing Solutions** — Students design evidence-based conservation networks using their biodiversity models to identify the most strategic connections between protected areas, prioritizing species at highest risk and enabling climate migration.
Disciplinary Core Idea	**LS2.C Ecosystem Dynamics, Functioning, and Resilience** — Biodiversity is the foundation of ecosystem stability and resilience. As species are lost, ecosystem functions degrade, and past critical thresholds, ecosystems can shift to fundamentally different and less productive states.
Crosscutting Concept	**Systems and System Models** — Students model biodiversity as a complex network of interacting species where the loss of individual nodes can cascade through connections to affect the entire system, and use this systems understanding to design effective conservation strategies.

CAST Testing Alignment

MULTIPLE CHOICE:

A region has lost 50% of its forest habitat. Which scenario results in higher extinction rates? A) The remaining 50% is in one continuous block B) The remaining 50% is in 20 small, isolated fragments C) The remaining 50% is in 20 fragments connected by wildlife corridors D) Fragmentation does not affect extinction rates if total area is the same

CONSTRUCTED RESPONSE:

A keystone pollinator species has been declining due to habitat fragmentation. Using your biodiversity model, predict what happens to the ecosystem when this species falls below its minimum viable population. Trace the extinction cascade through at least three levels of the species interaction network and explain why the total species impact exceeds the loss of just the pollinator.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Earth is currently experiencing its sixth mass extinction event — the first caused by a single species. The current extinction rate is estimated at 100-1,000 times higher than the natural background rate, rivaling the pace of the asteroid impact that ended the dinosaurs 66 million years ago. Unlike previous mass extinctions driven by catastrophic physical events, this one is driven by the cumulative effects of habitat destruction, fragmentation, climate change, pollution, and overexploitation — all consequences of human activity that are accelerating rather than abating.

The Scale of Biodiversity Loss

The 2019 Global Assessment Report on Biodiversity found that approximately 1 million animal and plant species are threatened with extinction within decades — more than at any time in human history. Since 1970, monitored wildlife populations have declined by an average of 69%. Amphibians are the most threatened vertebrate class, with 41% of species at risk. Insect populations have declined by 45% globally over the past 40 years, with devastating implications for pollination and food webs. The Living Planet Index tracks vertebrate populations and shows consistent, accelerating decline across all major ecosystems — freshwater species have declined by 83%, the most of any group. These losses are not just numbers — each represents the unraveling of ecological relationships built over millions of years of evolution.

Habitat Fragmentation: Death by a Thousand Cuts

Habitat fragmentation is arguably the most damaging driver of biodiversity loss because it multiplies the effects of all other threats. When continuous forest is divided by roads and agriculture, each resulting fragment suffers from edge effects — changes in microclimate, light, humidity, and wind penetration that alter habitat quality for 100-300 meters inward from each edge. A 100-hectare fragment may have only 20 hectares of true interior habitat. Small fragments cannot support minimum viable populations of large predators, dispersal between fragments is blocked for many species, and genetic isolation leads to inbreeding depression within a few generations. Critically, fragmentation also prevents species from migrating in response to climate change — they are trapped in habitat islands surrounded by hostile landscape.

Extinction Cascades: The Domino Effect

Ecosystems are networks of interdependent species, and the loss of one species can cascade through these connections in ways that multiply the direct impact. The classic example is the reintroduction of wolves to Yellowstone: wolves reduced elk overgrazing, which allowed willows and aspens to recover, which stabilized stream banks, which improved habitat for fish and beaver, which created ponds that supported amphibians and songbirds — one species affected dozens. The reverse is also true: losing a keystone species can trigger a cascade of declines. Pollinator loss threatens the 75% of food crops that depend on animal pollination. The decline of large herbivores reduces seed dispersal and nutrient cycling. The loss of apex predators causes herbivore population explosions that overgraze vegetation. Each lost species weakens the network for the survivors.

Conservation Corridors: Reconnecting the Landscape

The most promising strategy for combating fragmentation-driven extinction is creating wildlife corridors — strips of habitat that connect isolated protected areas. Corridors enable gene flow between isolated populations (preventing inbreeding), allow animals to migrate in response to climate change, increase effective habitat area by combining connected fragments, and enable recolonization after local extinctions. The Yellowstone to Yukon Conservation Initiative (Y2Y) is creating a 3,200-km corridor connecting protected areas from Yellowstone to the Canadian Yukon, enabling grizzly bear, caribou, and other wide-ranging species to maintain viable populations across their full range. Research consistently shows that connected reserves support 20-30% more species than isolated reserves of the same total area.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Land Use Change Rate, Climate Velocity, Habitat Connectivity, Population Viability, Species Interaction Network, Ecosystem Function Index, and Extinction Rate as key components of the biodiversity system.

E **Establish Relationships**

Students determine that habitat loss and fragmentation reduce population viability, that climate velocity creates migration pressure that fragmentation prevents, that species interaction networks transmit extinction cascades, and that connectivity is the key moderating variable.

V **Visualize & Model**

Students build a computational model showing how habitat loss and fragmentation interact with climate change to drive species decline, and how corridor connectivity can moderate extinction dynamics.

E **Evaluate Outcomes**

Students run continued habitat loss, climate-without-corridors, and connected landscape scenarios to test predictions about the relative effectiveness of different conservation strategies.

R **Revise & Extend**

Students add invasive species pressure, genetic diversity, or protected area effectiveness to explore how additional factors affect extinction dynamics and conservation outcomes.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"We\'re in the middle of the sixth mass extinction. Species are disappearing 1,000 times faster than normal. Today we\'re modeling why — and what we can do about it."*

Watch For

Show cover image. Quick stat: Since 1970, 69% of monitored wildlife populations have declined. Ask: What do you think is the main cause? Take predictions.

Slide 2: Learning Objectives

3 min

Say: *"By the end of today, you\'ll understand not just that species are disappearing, but the specific mechanisms that drive extinction — and why the pattern is accelerating."*

Watch For

Have students read objectives. Pre-teach vocabulary. Key concept: fragmentation — cutting habitat into pieces is worse than just losing area. Use the puzzle analogy: a complete picture with some pieces missing versus the same pieces scattered randomly.

Slide 3: Big Question

3 min

Say: *"The last time species disappeared this fast, an asteroid had just hit the planet. This time, the asteroid is us. But unlike an asteroid, we can choose to change course. What would that take?"*

Watch For

Quick-write: Students write their initial explanation for why extinctions are accelerating and what they think the most important intervention would be. Save for comparison.

Slide 4: LEVER Framework

2 min

Say: *"We\'ll model the extinction crisis as a system to understand the dynamics that drive it. This model will reveal something important: connectivity changes everything."*

Watch For

Preview LEVER steps. Emphasize that understanding the system dynamics of extinction helps us find the highest-leverage interventions — places where a small change produces a big result.

Slide 5: Activity 1: Components

8 min

Say: *"Which components are external pressures acting on biodiversity from outside, and which are internal responses within the ecological system?"*

Watch For

Guide sorting: External = Land Use Change Rate (human decisions), Climate Velocity (global warming). Internal = the five ecological responses. Discuss: Why is Habitat Connectivity internal even though humans build the roads that fragment it?

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the pathway from habitat loss to extinction. But also look for the multiplier effect — where does fragmentation make everything worse? Where do cascades propagate?"*

Watch For

Students draw arrows. Key discoveries: the cascading effect through Species Interaction Networks, and the central role of Habitat Connectivity in moderating multiple pathways. Ask: Which component, if improved, would have the biggest ripple effect?

Slide 7: Activity 3: Simulation

10 min

Say: *"Run all three scenarios and focus on the conservation corridor scenario. Does reconnecting habitat actually reduce extinction rates, even without stopping habitat loss?"*

Watch For

Students run scenarios. Key discussion: Compare the corridor scenario to continued habitat loss. The improvement from connectivity alone should be striking. Ask: Why does connecting the same total area of habitat produce such different results?

Slide 8: Discoveries

5 min

Say: *"Our model reveals four critical insights about why species are disappearing and what we can do. The most powerful finding: connectivity is a force multiplier for conservation."*

Watch For

Class discussion. Compare initial predictions to model evidence. Challenge: If corridors are so effective, why aren\'t we building them everywhere? What are the barriers?

Slide 9: STEM Challenge

12 min

Say: *"You have a limited budget to connect 15 protected areas in a developing region. Which connections do you prioritize? Use your model data to design the most effective network possible."*

Watch For

Groups design corridor networks on regional maps, prioritizing connections based on model evidence about species vulnerability, climate migration needs, and keystone species protection. Present and defend design choices.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Land Use Change Rate, Climate Velocity
- **INTERNAL:** Habitat Connectivity, Population Viability, Species Interaction Network, Ecosystem Function Index, Extinction Rate

Relationships

- **Land Use Change Rate to Habitat Connectivity:** NEGATIVE (-) — More rapid land conversion fragments remaining habitat into smaller, more isolated patches, reducing the corridors and connections that species need for movement, gene flow, and migration.
- **Habitat Connectivity to Population Viability:** POSITIVE (+) — Higher connectivity enables gene flow between populations, access to mates and resources across larger areas, and migration in response to climate change — all of which increase the likelihood that populations remain above minimum viable levels.
- **Climate Velocity to Population Viability:** NEGATIVE (-) — Faster climate zone shifts require faster species migration. When climate velocity exceeds a species\' dispersal ability — especially in fragmented landscapes — populations become trapped in unsuitable conditions and decline.
- **Population Viability to Species Interaction Network:** POSITIVE (+) — When populations are viable and stable, they maintain their roles in the species interaction network — as predators, pollinators, seed dispersers, and nutrient cyclers. Population decline weakens or removes these connections.
- **Species Interaction Network to Extinction Rate:** NEGATIVE (-) — A robust interaction network with many species and connections provides redundancy and resilience — the loss of one species can be partially compensated by others. As the network degrades, this resilience disappears and extinction cascades become more likely.
- **Ecosystem Function Index to Population Viability:** POSITIVE (+) — Healthy ecosystem function provides the resources, habitat quality, and environmental stability that species need to maintain viable populations. Declining function degrades habitat quality for all remaining species.

Key Discoveries

- Extinction rates accelerate non-linearly as habitat is lost — the first 30% of habitat loss causes relatively few extinctions, but beyond 70% loss, extinction rates explode because remaining populations are too small and isolated to be viable
- Habitat fragmentation is more damaging than equivalent area loss because it creates edge effects, blocks migration, isolates gene pools, and prevents species from tracking climate change — a single connected habitat supports far more species than the same area divided into isolated fragments
- Extinction cascades through Species Interaction Networks mean that the direct loss of one keystone species can trigger the decline of dozens of dependent species — the actual extinction toll is always higher than the directly impacted species count
- Wildlife corridors that reconnect fragmented habitats significantly improve Population Viability and reduce Extinction Rate even without stopping habitat loss — connectivity is a force multiplier that makes existing protected areas far more effective

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Extinction is natural, so the current rate is not a problem"	Extinction is indeed natural — the background rate is approximately 1-5 species per year over geological time. But the current rate is 100-1,000 times higher than this natural baseline, driven entirely by human activity. The last time species disappeared this fast was the asteroid impact 66 million years ago that killed 75% of all species. The difference between natural extinction and mass extinction is like the difference between a house settling over centuries and a house being demolished in a day — the outcomes are categorically different in scale and speed.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Protecting a few charismatic species (pandas, tigers) is sufficient conservation"	While flagship species generate public support and funding, focusing only on charismatic megafauna ignores the ecological reality that ecosystem function depends on thousands of less visible species. Insects pollinate 75% of food crops, fungi decompose organic matter and cycle nutrients, bacteria fix nitrogen that plants need, and countless invertebrates are the foundation of food webs. The loss of pollinators or soil organisms could be far more consequential for ecosystem function — and human food security — than the loss of any single large mammal. Conservation must protect entire functional ecosystems, not just photogenic species.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Technology and zoos can save species from extinction"	Captive breeding programs play a role in preventing imminent extinction for some species, but they cannot replace functional habitat. Most captive populations are too small for long-term genetic viability, behavioral adaptation to captivity reduces fitness in the wild, and reintroduction is impossible without suitable habitat. Furthermore, captive breeding is feasible for only a tiny fraction of threatened species — you cannot put a million species in zoos. The only way to prevent mass extinction at scale is to protect and reconnect habitat so that wild populations can sustain themselves. Technology is a supplement to habitat conservation, not a substitute for it.